Stochastic Signal Analysis

Chapter 3 Review Problem Set

Stationarity and Power Spectral Density

21 problems / 7 days / 3 problems per day

Based on the same eight verified past papers as the frequency report. Chapter 3 has 106 primary marks across this sample, averaging 13.25 marks per paper; the recent-four mean is 19.50.
All ten assigned and suggested homework exercises are included in full. Extra questions address joint WSS, delays, modulation spectra, white noise, and original-paper inconsistencies. Translations and adaptations are study materials, not official English textbook questions.

2026-10-27

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Use $R_{XY}(\tau) = \mathbb{E}[X(t+\tau)Y(t)]$ and the angular-frequency Fourier pair with inverse factor $1/(2\pi)$. Source labels identify every explicit adaptation. Each following page is one problem with working space.

Problem 1 — Day 1

Original recall and counterexample

Sections: 3.1.1

Stationarity: definitions and limits

State the definitions of strict-sense stationarity (SSS) and wide-sense stationarity (WSS), including the finite-second-moment requirement. Prove that SSS with finite second moments implies WSS, and that a jointly Gaussian WSS process is SSS. Does WSS imply SSS in general? Give a concrete counterexample with its time-dependent one-dimensional law and constant mean and correlation.

Working space

Problem 2 — Day 1

Textbook 3.1 / Optional homework

Sections: 3.1.1

Independent samples in continuous time

A random voltage signal $U(t)$ is statistically independent at distinct times. Its one-dimensional distribution is Gaussian with mean 0 and variance 2.

1. Find the density functions $f(u;t)$, $f(u_1,u_2;t_1,t_2)$ and $f(u_1,\ldots,u_k;t_1,\ldots,t_k)$, for arbitrary positive integer k.
2. Determine the stationarity of $U(t)$.

Working space

Problem 3 — Day 1

Textbook 3.4 / Required homework

Sections: 3.1.1

Independent stationary products

The random processes $X(t)$ and $Y(t)$ are independent and each is stationary. Prove that $Z(t) = X(t)Y(t)$ is also stationary.

Working space

Problem 4 — Day 2

Textbook 3.5 / Required homework

Sections: 3.1.1 / 3.3.1

Square-law output

Let $X(t) = 10\sin(\omega_0 t + \Theta)$, where ω_0 is a deterministic constant and Θ is uniformly distributed on $[-\pi, \pi]$. A square-law device produces $Y(t) = X^2(t)$. Find:

1. the mean of $Y(t)$;
2. its autocorrelation function;
3. whether $Y(t)$ is wide-sense stationary.

Working space

Problem 5 — Day 2

Textbook 3.9 / Required homework

Sections: 3.1.2 / 3.3.1

Sums, differences and cross-correlation

The stationary random processes $X(t)$ and $Y(t)$ are statistically independent, have zero means, and have autocorrelation functions

$$R_X(\tau) = \mathrm{e}^{-|\tau|}, \qquad R_Y(\tau) = \cos(2\pi\tau).$$

1. Find the autocorrelation of $Z(t) = X(t) + Y(t)$.
2. Find the autocorrelation of $W(t) = X(t) - Y(t)$.
3. Find the cross-correlation $R_{ZW}(\tau)$.

Working space

Problem 6 — Day 2

Adapted from 202312 Q1; finite-power correction

Sections: 3.1.1 / 3.1.2 / 3.4.2

Joint WSS and rational spectra

Let X and Y be independent zero-mean real WSS processes with

$$S_X(\omega) = \frac{16}{\omega^2 + 16}, \qquad S_Y(\omega) = \frac{8}{\omega^2 + 4}.$$

Put $Z = X + Y$ and $V = X - Y$. Prove that X, Y and also Z, V are jointly WSS. Find S_Z, R_{XZ} and R_{VZ}. State the cross-correlation convention you use.
The original paper uses $S_Y = \omega^2/(\omega^2 + 16)$, which is not integrable. This problem explicitly replaces it by a finite-power spectrum; Problem 21 diagnoses the original.

Working space

Problem 7 — Day 3

Adapted from 202112 Q2(a–d)

Sections: 3.1.1 / 3.1.2 / 3.3.1

Delayed radar echo

Let $X(t)$ and $N(t)$ be independent zero-mean real WSS processes with correlations R_X and R_N. Let $Y(t) = \alpha X(t-d)+N(t)$, where α and d are deterministic real constants. Using $R_{XY}(\tau) = \mathbb{E}[X(t+\tau)Y(t)]$, find the mean and autocorrelation of Y, both R_{XY} and R_{YX}, and establish the WSS of Y and the joint WSS of X, Y. Explain why the delay signs differ in the two cross-correlations.

Working space

Problem 8 — Day 3

Textbook 3.7 / Optional homework

Sections: 3.2.2

Cyclostationary shifts

The real random signal $Y(t)$ is wide-sense cyclostationary with cycle period 100. Its mean satisfies $m(10) = 20$, and its two-time correlation satisfies $R(5, 1) = 10$. Find:

1. $\mathbb{E}[5Y(110)]$ and $\mathbb{E}[10Y(310) + 50]$;
2. $\mathbb{E}[Y(105)Y(101)]$ and $\mathbb{E}[30Y(205)Y(201) + 200]$;
3. $\mathbb{E}[10Y(305)Y(301) + 6Y(210) + 80]$.

Working space

Problem 9 — Day 3

Textbook 3.14 / Required homework

Sections: 3.3.1 / 3.4.2

Admissible autocorrelations

Two zero-mean wide-sense stationary random processes $X(t)$ and $Y(t)$ have variances $\sigma_X^2 = 5$ and $\sigma_Y^2 = 10$. Can the following functions be their autocorrelation functions? Explain each answer. Here $u(\tau)$ denotes the unit step.

1. $R_X(\tau) = 5u(\tau)\exp(-3\tau)$.
2. $R_X(\tau) = 5\sin(5\tau)$.
3. $R_Y(\tau) = 9/(1+2\tau^2)$.
4. $R_Y(\tau) = -\cos(6\tau)\exp(-|\tau|)$.
5. $R_X(\tau) = 5[\sin(3\tau)/(3\tau)]^2$.
6. $R_Y(\tau) = 6 + 4\sin(10\tau)/(10\tau)$.
7. $R_X(\tau) = 5\exp(-|\tau|)$.
8. $R_Y(\tau) = 6 + 4\exp(-3\tau^2)$.

Removable singularities are understood by their limits.

Working space

Problem 10 — Day 4

Textbook 3.16 / Required homework

Sections: 3.1.1 / 3.3.1

A shared random amplitude

The processes $X(t)$ and $Y(t)$ are independent and each is stationary, with

$$R_X(\tau) = 2\mathrm{e}^{-|\tau|}\cos(\omega_0\tau), \qquad R_Y(\tau) = 9 + \exp(-3\tau^2).$$

Let $Z(t) = AX(t)Y(t)$, where the random variable A has mean 2 and variance 9, and A, $X(t)$ and $Y(t)$ are mutually independent. Find the mean, variance and autocorrelation of $Z(t)$.

Working space

Problem 11 — Day 4

Textbook 3.21 / Required homework

Sections: 3.4.2

Real-signal PSD admissibility

Which of the following functions can be power spectral densities of real random signals? Explain why or why not.

1. $[\sin \omega/\omega]^2$.
2. $\omega^2/(\omega^6 + 3\omega^2 + 3)$.
3. $\omega^2/(\omega^4 - 1) - \delta(\omega)$.
4. $\omega^4/(\mathrm{j}\omega^6 + \omega^2 + 1)$.
5. $|\omega|/(\omega^4 + 2\omega^2 + 1)$.
6. $\exp[-(\omega - 1)^2]$.

Working space

Problem 12 — Day 4

Original transform-pair exercise

Sections: 3.3.1 / 3.3.2 / 3.4.2

A spectrum with continuous and line parts

A zero-mean real WSS signal has

$$R_X(\tau) = 2\mathrm{e}^{-3|\tau|} + 4\cos(5\tau) + 1.$$

Find its PSD and mean-square value. Explain why a zero-mean signal can have a zero-frequency spectral impulse. Construct it as a sum of three independent zero-mean WSS components, specifying the correlation of each component; no stochastic differential equation is required.

Working space

Problem 13 — Day 5

Textbook 3.25 / Required homework

Sections: 3.4.2

Causal-looking cross-correlation and cross-spectrum

Two random processes $X(t)$ and $Y(t)$ are jointly stationary, with cross-correlation

$$R_{XY}(\tau) = \begin{cases} 9\mathrm{e}^{-3\tau}, & \tau \geq 0, \\ 0, & \tau < 0. \end{cases}$$

Find the cross-power spectral densities $S_{XY}(\omega)$ and $S_{YX}(\omega)$.

Working space

Problem 14 — Day 5

Textbook 3.19 / Optional homework

Sections: 3.4.2

Inverse transforms and infinite-power boundary

Find the autocorrelation function and the mean-square value of a stationary signal $X(t)$ for each of the following power spectral densities:

1. $S_X(\omega) = \omega^2/(\omega^4 + 3\omega^2 + 2)$.
2. $$S_X(\omega) = \begin{cases} 8\delta(\omega) + 20(1 - |\omega|/10), & |\omega| \le 10, \\ 0, & |\omega| > 10. \end{cases}$$
3. $S_X(\omega) = p\cos^4(\omega\tau_0)$, where p is a constant.

Working space

Problem 15 — Day 5

Adapted from 202212 Q2(a–c)

Sections: 3.1.1 / 3.4.2

Random-phase multiplicative modulation

Let $\xi(t)$ be a real WSS process with finite second moments, mean m_ξ, correlation R_ξ and PSD S_ξ. Let $\Theta \sim U[0, 2\pi)$ be independent of the entire process. For $\omega_0 > 0$, put

$$\eta(t) = \xi(t)\cos(\omega_0 t + \Theta).$$

Derive m_η, R_η and S_η, and prove WSS. Do not assume $m_\xi = 0$. Explain how any DC spectral line of ξ moves under this modulation.

Working space

Problem 16 — Day 6

Adapted from 202512 Q2(a,b); consistent amplitude model

Sections: 3.1.1 / 3.4.2

Nonuniform phase: what must vanish?

Let $A(t)$ be a real WSS process with mean 1 and covariance $C_A(\tau) = \mathrm{e}^{-|\tau|}$. A phase Θ is independent of the entire process and has characteristic function $\phi_\Theta(v) = \mathbb{E}[\mathrm{e}^{\mathrm{j}v\Theta}]$. For $\omega_0 > 0$, put $Z(t) = A(t)\cos(\omega_0 t + \Theta)$.

1. Derive its mean and two-time correlation, and give necessary and sufficient conditions on $\phi_\Theta(1), \phi_\Theta(2)$ for WSS.
2. Under these conditions find S_Z.
3. If X is an independent real WSS process, find S_{X+Z}.

The covariance, rather than uncentered correlation, is specified here to avoid the inconsistency in the original paper's part (c).

Working space

Problem 17 — Day 6

Adapted from 202412 Q2(a,b); zero means and nonzero carrier specified

Sections: 3.1.1 / 3.1.2 / 3.4.2

Quadrature modulation and joint stationarity

Let X, Y be jointly WSS real processes with zero means. Let $\omega_0 > 0$, and define

$$V(t) = X(t)\cos(\omega_0 t) + Y(t)\sin(\omega_0 t).$$

Expand $R_V(t+\tau, t)$ and show that WSS is equivalent to

$$R_X(\tau) = R_Y(\tau) = R_0(\tau), \qquad R_{XY}(\tau) = -R_{YX}(\tau).$$

Under these conditions derive $R_V(\tau)$ and $S_V(\omega)$ in terms of S_0, S_{XY}. Simplify further if X and Y are uncorrelated at every lag. Use $R_{XY}(\tau) = \mathbb{E}[X(t+\tau)Y(t)]$.

Working space

Problem 18 — Day 6

Adapted from 202412 Q1(a–c)

Sections: 3.1.1 / 3.1.2

Sampling at an integer number of cycles

For integer n, let $X[n] = \sin(2\pi n + \Phi)$ and $Y[n] = \cos(2\pi n + \Phi)$, with $\Phi \sim U(0, \pi)$. Find both means, both autocorrelations, and their cross-correlation. Are the sequences orthogonal, uncorrelated, independent, jointly WSS and jointly SSS? Justify each statement. Explain why continuous-time uniform-full-phase shortcuts do not apply.

Working space

Problem 19 — Day 7

Adapted from 202112 Q4(a); carrier separation stated

Sections: 3.3.2 / 3.4.2

Power from a bandpass cosine spectrum

A zero-mean real WSS signal has the two-sided angular-frequency PSD

$$S_X(\omega) = \begin{cases} P\cos[\pi(\omega-\omega_0)/\Delta], & |\omega-\omega_0| \le \Delta/2, \\ P\cos[\pi(\omega+\omega_0)/\Delta], & |\omega+\omega_0| \le \Delta/2, \\ 0, & \text{otherwise}, \end{cases}$$

where $P > 0$, $\Delta > 0$ and $\omega_0 > \Delta/2$. Sketch the spectrum with its peak heights and band edges. Determine the mean-square value and variance, showing both lobe areas and the $1/(2\pi)$ factor. Here P labels the spectral height, not the total power.

Working space

Problem 20 — Day 7

Original exercise; tools used in 202104 Q2 and recent RC-noise questions

Sections: 3.3.2 / 3.4.2 / 3.5.1

White noise, finite bandwidth and finite variance

Ideal continuous-time white noise has two-sided angular-frequency PSD $S_N(\omega) = N_0/2$, with $N_0 > 0$. A band-limited approximation has

$$S_{N_B}(\omega) = \frac{N_0}{2}\mathbf{1}_{\{|\omega|\le B\}}, \qquad B > 0.$$

Find $R_{N_B}(\tau)$ and its mean-square value, and take the distributional limit $B \to \infty$. Explain why ideal continuous-time white noise cannot simultaneously have this nonzero flat PSD and a finite pointwise variance. Does white imply Gaussian? Contrast this model with the finite-variance independent-sample process in textbook 3.1.

Working space

Problem 21 — Day 7

Original diagnostic based on 202312 Q1 and 202512 Q2(c)

Sections: 3.1.1 / 3.3.2 / 3.4.2

Diagnose two original-paper inconsistencies

1. The 202312 paper gives $S_Y(\omega) = \omega^2/(\omega^2 + 16)$. Find its formal inverse Fourier transform. Can an ordinary finite-variance WSS signal have this PSD? Explain without assigning a numeric value to $\delta(0)$.
2. The 202512 paper states $m_A \neq 0$ and later gives the uncentered correlation $R_A(\tau) = \mathrm{e}^{-|\tau|}$. Explain why these cannot both describe a real WSS process with finite second moments.
3. Give a consistent way to reinterpret or modify each model. Separate these explicit changes from the original questions.

Working space